Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	Bone Strain During 0G Exercise		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	03/27/2001	**End Date:**	12/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	n/a		
Key Personnel Changes/Previous PI:	n/a		
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	Considerable NASA resources have been devoted to a vibration-isolated treadmill (TVIS) that has been installed in the International Space Station (ISS). This device is attractive because programs may be designed to allow simultaneous exercise countermeasures to be applied against bone, muscle, and cardiovascular adaptations. Despite the need for on-orbit exercise using TVIS, almost no research has been conducted to determine the acceptability of the subject load device (SLD) and harness used to tether crewmembers to the treadmill. Data regarding the bone strains produced by exercise countermeasures, such as running in reduced gravity, are also unavailable, yet there is little doubt that bone strain plays an important intermediary role in the signal transduction pathway responsible for bone maintenance and functional adaptation. This study in simulated microgravity was designed to evaluate the comfort and function of a new SLD design applied to subjects running in a system that allows controlled distribution of the load between the waist and		

Task Description:	shoulders during treadmill running. Using the Zero Gravity Locomotion Simulator (ZLS), twelve subjects completed nine, five minute running trials at a constant speed (3.35 ms-1) using 3 SLD loads (50%, 75% and 100% BW) and 3 shoulder-to-waist loading ratios (50:50, 60:40 and 70:30) presented at random. Subjective ratings of discomfort, ground reaction forces and SLD loads were collected. 100% BW in the ZLS resulted in similar GRF profiles to those obtained from the literature for overground running. The full body weight loading conditions were comfortably tolerated and a shoulder-to-waist load ratio of 70:30 minimized discomfort at the waist. The ZLS and its new SLD system are promising methods of simulating overground running in zero gravity as evidenced by the low discomfort levels of subjects and their ground reaction force profiles. The new SLD system still shows variation around the set level and the investigators will be exploring new approaches to applying loads to the harness worn by subjects running in the ZLS. The second phase of this investigation will examine the feasibility of the new system in a long-term exercise program, such as the 3 to 6 months that crewmembers are currently spending on the ISS. During the third phase, ZLS data will then be used as input to robotic cadaver simulations in order to directly measure bone strains in cadaver limbs moving through accurate kinematic and kinetic reconstructions of orbit-based locomotion. Thus, these sequential experiments will provide the unique psychophysical and biomechanical information needed to design better exercise countermeasures to bone loss on the ISS. They will also provide fundamental new data on the relationship between external loading and strain in human bone.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research aim of this study is to provide psychophysiological and biomechanical information to design better exercise programs to prevent bone loss.
Task Progress:	In the last 12 months, the ZLS has been installed in the new Biomechanics Laboratiory located within the Department of Biomedical Engineering at the Cleveland Clinic Foundation. During the relocation and construction, the entire ZLS system received significant upgrades and refinements. A new treadmill motor and belt have been installed, a complete redesign of the SLD system is underway, including new linear motors, load splitters and harness, a new roping system and bungee cords for subject suspension and various other hardware upgrades. The ZLS upgrades are nearing completion and the team anticipates proceeding with the experiments outlined in this protocol shortly. This experiment has been presented at Grand Rounds at Johnson Space Center and to the Space Station Program Control Board (SSPCB). This project is currently being considered for a no-cost extension.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Significant Media Coverage	NBC Medical Miracles "New Frontiers in Research: Bone Density, Stem Cells, Macular Degeneration" . NBC Medical Miracles , Apr-2004